INCH-POUND

MIL-M-38510/657
AMENDMENT 5
6 December 1994
SUPERSEDING
MIL-M-38510/657
AMENDMENT 4
22 August 1994

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS,
BUFFER GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/657, dated 30 September 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

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★ TABLE I, Additional supply current quiescent per input pin: Delete in its entirety and substitute as follows:

"

Additional supply current quiescent per input pin (one unit load)	$I_{CC\Delta}$ 7/	For all inputs, V_{IL} = 0.4 V and V_{IH} = 2.4 V Test pin at V_{IN} = 2.4 V When not being tested, control pins at V_{IH} = V_{CC} and V_{IL} = GND I_O = 0 µA	52,54	5.5 V		1.50	mA
			51			2.20	
			53,55 60,61			2.50	

"

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TABLE I, Output enable time, device types 06, 07, 08, 09, 60 and 61: Delete in its entirety and substitute as follows:

"

Output enable time 5/, 6/	t_{PZH}	C_L = 50 pF minimum R_L = 1 kΩ ±10%	06,07	4.5 V	5	51	ns
			08,09	4.5 V	5	44	ns
			60,61	4.5 V	7	53	ns
	t_{PZL}		06,07	4.5 V	5	51	ns
			08,09	4.5 V	5	44	ns
			60,61	4.5 V	10	53	ns

"

TABLE I, Transition: Delete in its entirety and substitute as follows:

"

Transition 5/, 6/	t_{THL} t_{TLH}	C_L = 50 pF ±10%	01,02	4.5 V	3	20	ns
			03,04, 06-11	4.5 V	2	16	ns
			05	4.5 V	1	16	ns
			51,52	4.5 V	3	20	ns
			53-55, 60,61	4.5 V	2	16	ns

"

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TABLE III, device type 05, t_{TLH}, T_C = +25°C in the max limits column, delete "2" and substitute "1".
TABLE III, device type 05, t_{TLH}, T_C = +125°C in the max limits column, delete "2" and substitute "1".
TABLE III, device type 05, t_{TLH}, T_C = -55°C in the max limits column, delete "2" and substitute "1".

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* TABLE III, device type 51, $I_{CC\Delta}$, test number 15: Delete in its entirety and substitute as follows:

"

I_{CC8}	215	2.4V		0.4V		0.4V		GND		0.4V		0.4V		0.4V	5.5V	V_{CC}		1.70	1.50	2.20	mA
	216	0.4V		2.4V		"		"		"		"		"	"	"		"	"	"	"
	217	"		0.4V		2.4V		"		"		"		"	"	"		"	"	"	"
	218	"		"		0.4V		"		2.4V		"		"	"	"		"	"	"	"
	219	"		"		"		"		0.4V		2.4V		"	"	"		"	"	"	"
	220	"		"		"		"		"		0.4V		2.4V	"	"		"	"	"	"

"

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TABLE III, device type 53, $I_{CC\Delta}$, test numbers 23 and 24: Delete in their entirety and substitute as follows:

"

I_{CC8}	223	2.4V	0.4V		0.4V		0.4V		GND	0.4V		0.4V		0.4V		GND	5.5V	V_{CC}		2.20	1.80	2.50	mA
	224	GND	2.4V		"		"		"	"		"		"		"	"	"		"	"	"	"
	225	"	0.4V		2.4V		"		"	"		"		"		"	"	"		"	"	"	"
	226	"	"		0.4V		2.4V		"	"		"		"		"	"	"		"	"	"	"
	227	"	"		"		0.4V		2.4V	"		"		"		"	"	"		"	"	"	"
	228	"	"		"		"		0.4V	"		2.4V		"		"	"	"		"	"	"	"
	229	"	"		"		"		"	0.4V		2.4V		"		"	"	"		"	"	"	"
	230	"	"		"		"		"	"		0.4V		2.4V		"	"	"		"	"	"	"
	231	"	"		"		"		"	"		"		0.4V		2.4V	"	"		"	"	"	"
	232	"	"		"		"		"	"		"		"		0.4V	2.4V	"		"	"	"	"

"

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TABLE III, device type 55, $I_{CC\Delta}$, test numbers 23 and 24: Delete in their entirety and substitute as follows:

"

I_{CC8}	223	2.4V	0.4V		0.4V		0.4V		GND	0.4V		0.4V		0.4V		GND	5.5V	V_{CC}		2.20	1.80	2.50	mA
	224	GND	2.4V		"		"		"	"		"		"		"	"	"		"	"	"	"
	225	"	0.4V		2.4V		"		"	"		"		"		"	"	"		"	"	"	"
	226	"	"		0.4V		2.4V		"	"		"		"		"	"	"		"	"	"	"
	227	"	"		"		0.4V		2.4V	"		"		"		"	"	"		"	"	"	"
	228	"	"		"		"		0.4V	"		2.4V		"		"	"	"		"	"	"	"
	229	"	"		"		"		"	0.4V		2.4V		"		"	"	"		"	"	"	"
	230	"	"		"		"		"	"		0.4V		2.4V		"	"	"		"	"	"	"
	231	"	"		"		"		"	"		"		0.4V		2.4V	"	"		"	"	"	"
	232	"	"		"		"		"	"		"		"		0.4V	2.4V	"		"	"	"	"

"

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TABLE III, device type 60, $I_{CC\Delta}$, test numbers 23 and 24: Delete in their entirety and substitute as follows:

I_{CC8}		223	2.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	GND								GND	5.5V	V_{CC}		2.30		1.5		2.50 mA
		224	GND	2.4V	"	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		225	"	0.4V	2.4V	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		226	"	"	0.4V	2.4V	"	"	"	"	"	"								"	"	"		"		"		"	"
		227	"	"	"	0.4V	2.4V	"	"	"	"	"								"	"	"		"		"		"	"
		228	"	"	"	"	0.4V	2.4V	"	"	"	"								"	"	"		"		"		"	"
		229	"	"	"	"	"	0.4V	2.4V	"	"	"								"	"	"		"		"		"	"
		230	"	"	"	"	"	"	0.4V	2.4V	"	"								"	"	"		"		"		"	"
		231	"	"	"	"	"	"	"	0.4V	2.4V	"								"	"	"		"		"		"	"
		232	"	"	"	"	"	"	"	"	0.4V	"								2.4V	"	"		"		"		"	"

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TABLE III, device type 60, t_{PZH}, test number 133, test limits, T_C = -55°C, min limits column, change from """ to "7".

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TABLE III, device type 60, t_{PZH}, test number 141, test limits, T_C = -55°C, min limits column, change from "10" to "7".

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TABLE III, device type 61, $I_{CC\Delta}$, test numbers 23 and 24: Delete in their entirety and substitute as follows:

I_{CC8}		223	2.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	GND								GND	5.5V	V_{CC}		1.90		1.5		2.50 mA
		224	GND	2.4V	"	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		225	"	0.4V	2.4V	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		226	"	"	0.4V	2.4V	"	"	"	"	"	"								"	"	"		"		"		"	"
		227	"	"	"	0.4V	2.4V	"	"	"	"	"								"	"	"		"		"		"	"
		228	"	"	"	"	0.4V	2.4V	"	"	"	"								"	"	"		"		"		"	"
		229	"	"	"	"	"	0.4V	2.4V	"	"	"								"	"	"		"		"		"	"
		230	"	"	"	"	"	"	0.4V	2.4V	"	"								"	"	"		"		"		"	"
		231	"	"	"	"	"	"	"	0.4V	2.4V	"								"	"	"		"		"		"	"
		232	"	"	"	"	"	"	"	"	0.4V	"								2.4V	"	"		"		"		"	"

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TABLE III, device type 61, t_{PZH}, test number 133, test limits, T_C = -55°C, min limits column, change from """ to "7".

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TABLE III, device type 61, t_{PZH}, test number 141, test limits, T_C = -55°C, min limits column, change from "10" to "7".

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Air Force - 19, 85, 99
Navy - AS, CG, MC, OC, SH

Preparing activity:
DLA - ES

(Project 5962-1412)

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